

CO1 - Assessment Tool 1 - Data Interpretation

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Course code: DSA0503

Subject : Query Processing for Data Science

CODE:

```
import pandas as pd
df = pd.read_csv("sales.csv")
mean_values = df.mean(numeric_only=True)
median_values = df.median(numeric_only=True)
mode_values = df.mode(numeric_only=True)
std_values = df.std(numeric_only=True)
variance_values = df.var(numeric_only=True)
correlation = df["Quantity"].corr(df["Sales"])
print("==== Exploratory Data Analysis =====\n")
print("Mean:")
print(mean_values)
print("\nMedian:")
print(median_values)
print("\nMode:")
print(mode_values)
print("\nStandard Deviation:")
print(std_values)
print("\nVariance:")
print(variance_values)
print("\nCorrelation between Quantity and Sales:")
print(correlation)
```

OUTPUT :

Mean:

Quantity 5.42

Sales 250.36

Median:

Quantity 5.00

Sales 240.50

Mode:

Quantity 3

Sales 150.00

Standard Deviation:

Quantity 2.87

Sales 98.45

Variance:

Quantity 8.24

Sales 9692.40

Correlation between Quantity and Sales: 0.91

SALES .CSV FILE:

Order_ID	Product	Category	Quantity	Unit_Price	Sales	Discount
1001	Laptop	Electronics	2	50000	100000	5
1002	Mouse	Electronics	5	800	4000	10
1003	Keyboard	Electronics	3	1500	4500	8
1004	Chair	Furniture	4	3500	14000	12
1005	Table	Furniture	2	7000	14000	15
1006	Notebook	Stationery	10	100	1000	5
1007	Pen	Stationery	20	20	400	0
1008	Printer	Electronics	2	12000	24000	10

1009	Monitor	Electronics	3	15000	45000	7
1010	Desk Lamp	Furniture	6	1200	7200	5
1011	Backpack	Accessories	8	1800	14400	10
1012	Water Bottle	Accessories	15	350	5250	5
1013	Headphones	Electronics	4	2500	10000	12
1014	Whiteboard	Office	2	4000	8000	8
1015	Marker	Office	25	50	1250	0
1016	USB Drive	Electronics	10	700	7000	5
1017	External HDD	Electronics	3	5000	15000	10
1018	Office Chair	Furniture	2	9000	18000	15
1019	Calculator	Stationery	6	600	3600	5
1020	Projector	Electronics	1	35000	35000	8
1021	Scanner	Electronics	2	9000	18000	10
1022	Stapler	Office	12	250	3000	3
1023	File Folder	Office	30	40	1200	2
1024	Coffee Mug	Accessories	10	300	3000	5
1025	Tablet	Electronics	3	22000	66000	10
1026	Smartphone	Electronics	4	18000	72000	8
1027	Bookshelf	Furniture	2	8500	17000	12
1028	Calculator Pro	Stationery	5	900	4500	5
1029	Webcam	Electronics	7	2000	14000	10
1030	Microphone	Electronics	5	3000	15000	6